

Spring 2023
Transport Processes (Unit Operations)
CHE 354: #15435

COURSE DESCRIPTION

INSTRUCTOR: J. Steven Swinnea, Chemical Engineering/Texas Materials Institute
OFFICE: ETC 9.114
PHONE: 512-471-3173
EMAIL: swinnea@che.utexas.edu
Please begin subject line with CHE354: for faster response. I get lots of junk mail and this will help me find your message faster.

LECTURES: Monday, Wednesday and Friday; 12:00 PM until 1:00 PM CPE 2.216
RECITATION Wednesday 1:00 PM until 2:00 PM CPE 2.218
OFFICE HOURS: Mondays & Wednesdays; 3 PM -4:30 PM, Thursdays 9AM - 10:30 AM (and by appointment)

COURSE OBJECTIVES:

The course covers analysis and design of heat transfer equipment, fluid flow equipment and some interphase contacting devices. This course serves to integrate the fundamental concepts learned in ChE 319/353, 210 and 317 with practical chemical engineering problems and commercial applications. Key objectives of this course are to enhance problem definition and problem solving skills of the student, and to improve the student's ability to make calculations and obtain correct final answers. Process safety concepts are also introduced. Students should begin developing a tool-kit for common commercial and industrial problems..

WHAT SHOULD I ALREADY KNOW

- The catalog lists ChE 319/353 as a prerequisite. The course relies heavily on the mass, momentum, heat and energy balance equations developed in 319/353, 210 and 317. Some spreadsheet experience from ChE 210 will be useful for some problems.
- Familiarity with units and conversion from one unit system to another is needed. Ability to work exclusively in assigned unit systems (SI, CGS, US Customary) without converting to other systems is assumed.
- Ability to perform macroscopic mass and energy balances under steady state and dynamic conditions (from ChE 210, ChE 317 and 319/353).
- Familiarity with equations of state, particularly the ideal gas law (from ChE 317).
- Familiarity with concepts of enthalpy, work, heat and other thermodynamic quantities (from ChE 317).
- Knowledge of the fundamentals of transport phenomena including familiarity with the equations used to describe momentum and heat transfer, an understanding of dimensionless groups, and the ability to derive momentum and energy balance equations (from ChE 319/353).

MAIN KNOWLEDGE AND SKILLS DEVELOPED

- Familiarity with the methods used to analyze and design fluid flow and fluid processing equipment, heat transfer equipment, and some interphase contacting devices as well as some of the operating characteristics.
- Recognition of the challenges of designing processing equipment for compressible fluids and the ability to use the governing equations in flow and processing applications.
- The ability to analyze and size piping networks, valves, pumps, compressors and flow meters.
- The ability to determine heat transfer coefficients and the ability to size and rate heat exchangers.
- The ability to apply the principles of momentum and heat transport to the analysis of a wide range of processing equipment including evaporators, separation equipment, fluidized beds, and chemical reactors.
- Introduction to the topic of safety in the chemical process industries including topics related to process equipment design and operation.

ABET UNDERGRADUATE PROGRAM OUTCOMES:

The objective of the chemical engineering degree program is to prepare students for professional practice after they earn the bachelor's degree or an advanced degree. While many of the program outcomes are touched on in ChE 354, the major identified outcomes are:

- An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- An ability to apply engineering design to produce solutions that meet specified needs with consideration for public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

Planned Course Major Topics

1. Dimensional Analysis
2. Static fluid phenomena
3. Flow of incompressible fluids
4. Flow of compressible fluids
5. Design of fluid flow equipment
6. Flow past objects
7. Conductive heat transfer
8. Convective heat transfer
9. Radiation heat transfer
10. Heat transfer equipment design

HOW TO SUCCEED IN THIS COURSE

- **Focus on concepts.** This course is primarily about concepts. We will use mathematics to represent important concepts in process engineering equipment. Equation sheets will be provided during midterms and final exams. While you are not required to memorize all formulas and equations, you should be able to recognize important equations, explain what they describe, and the factors appearing in them. Additionally, you should be able to explain the underpinning assumptions and applicability of formulas describing physical phenomena;
- **Be reflective.** The assigned homework provides an effective way for practicing the concepts covered in class. The assignments will also give students the opportunity to reflect more deeply on your own and others' perspectives and assumptions, and be expected to draw, articulate, and justify conclusions with solid reasons.
- **Participate.** Learning is a collaborative process and thinking is not a spectator sport. Participate in class by communicating your understanding and testing others' understanding with questions and dialogue. The more you put into it, the more you will get out of it. Active participation includes being prepared to discuss readings, assignments, and concepts, engaging yourself in classroom activities and discussion, and putting your best effort in both formal and informal assignments. Regardless of the format we use to conduct class sessions, consider your participation to be the equivalent to a face-to-face class session and be prepared to engage actively and thoughtfully

COURSE REQUIREMENTS

How Will We Learn?

Attendance in lecture or discussions is encouraged but not required, but see Quizzes below. If you do attend class, please arrive promptly, attentive and ready to work.

- Lecture will be used to introduce new material and provide examples. Short unannounced quizzes may be given occasionally to test preparedness and understanding. They will come mostly from reading assignments.
- Recitation will be used to conduct short concept quizzes, provide homework help and solutions, and to devote extra time to difficult topics.
- Reading Assignments will be listed on Canvas. Read the assignments and work example problems in your textbook. Have a good understanding of the concepts.
- Canvas will be used to post homework assignments and solutions, assignment grades and any announcements. Please check Canvas often.

REQUIRED MATERIALS

McCabe, Smith, Harriott, *Unit Operations of Chemical Engineering*, 7th Edition. McGraw-Hill, 2005.

REUUIRED DEVICES/SOFTWARE

A good scientific calculator will be useful for completing homework, quizzes and exams. I was introduced to the TI nSpire CX CAS some time ago. It is pretty awesome and cost less than my first ever calculator (a **1974 TI-50**). Chemical Engineering students of course are required to have a laptop at their disposal and there will be some assignments that require spreadsheet solutions.

ASSESSMENT

GRADING FOR THIS COURSE:

Course grades will be calculated using scores from three in-class exams, a final exam, homework assignments and quizzes/class participation, with weights given in the table below on the left. Letter grades will be assigned based on these scores, as given in the table below. Plus/minus designations will be used for grades near the boundaries of the letter-grade transitions shown in the table.

Item	Weight
Exam I	19%
Exam II	19%
Exam III	19%
Exam IV	19%
Homework	16%
Quizzes/Classwork	8%

<59	60-63	64-66	67-69	70-73	74-76	77-79	80-83	84-86	87-89	90-93	94-100
F	D-	D	D+	C-	C	C+	B-	B	B+	A-	A

HOMEWORK:

Homework assignments will be due approximately once per week. At the assigned due date/time through Canvas. Late homework assignments will not be accepted! Students are encouraged to discuss the homework exercises and work together to understand solution strategies, but the solutions turned in by each student must be the final work of the individual student. Homework assignments must be properly labeled and completed in a neat, professional manner, please use 317 standards. Nonconforming assignments may be returned without credit. Each page or image of a page must have your name, EID and CHE354 as a heading. The assignment should be uploaded as a single PDF file. For word processed material, you should print directly to PDF.

EXAMS:

Exams, except the final exam, will be conducted during the normal lecture/recitation period on the tentative dates specified. If the exam date needs to be modified due to schedule disruption, you will be notified well in advance. In addition to an in-class portion, exams may also include some "take-home" problems. The final will be administered at the time assigned by the University. Students are allowed to use their textbook and lecture notes on the exams. Students are also allowed to use a computer, a calculator, pencils, pens, straight edges and drafting triangles. No communication with any other person is allowed. No other resources are allowed. Using resources other than those allowed will invoke academic dishonesty procedures.

Quizzes:

Very short quizzes on current class material may be given in both lecture and recitation. There will be no makeup option for these. One or two of these will be dropped to account for occasional absences.

REGRADES:

Errors in grading should be noted *in writing* and provided to the instructor with the graded item for review *no later than one week* after the item is made available to the student. *No regrades will be accepted without a written description* of the grading error, and regrades will only be conducted by the instructor.

ATTENDANCE & CLASS PARTICIPATION:

Attendance and participation in lecture are required and will be considered in grading. Students are expected to keep up with the assigned reading and be prepared to answer questions on these readings during lecture. The instructor must be given written notice a minimum of two weeks in advance for any absences other than emergencies. Refer to the General Information Catalog, available on-line, for University policy on attendance. If

absence is required because of an emergency, such as personal illness or severe illness of a family member, the instructor should be notified at the earliest convenient opportunity.

IMPORTANT DATES:

<i>Fourth Class Day:</i>	January 12, 2023
<i>Twelfth Class Day:</i>	January 25, 2023
<i>Engineering Expo:</i>	February 3, 2023 10AM – 4PM (Virtual)
<i>First exam (tentative date*):</i>	February 15, 2023
<i>Second exam (tentative date*):</i>	March 22, 2023
<i>Third exam (tentative date*):</i>	April 19, 2023

(* Exams will be scheduled to correspond to appropriate break points in the subject matter. You will be informed of the exact exam date no later than one week before the exam. Let's stick to the schedule!)

Final exam: April 28th, 2023; 1:00 PM until 3:00 PM

The final exam must be taken at the time and on the date given in the *Course Schedule*. You may use a calculator (or a slide rule), and a straight edge may be useful.

COURSE OUTLINE AND REQUIRED READINGS FROM THE COURSE TEXT:

All instructions, assignments, readings, and essential information will be on the Canvas website at <https://utexas.instructure.com>. Check this site regularly and use it to ask questions about the course schedule.

Changes to the schedule may be made at my discretion and if circumstances require. It is your responsibility to note these changes when announced (although I will do my best to ensure that you receive the changes with as much advanced notice as possible).

COURSE POLICIES

LATE WORK:

Late work will be handled on a case-by-case basis. Generally, except in the case of an excused absence, work submitted after an assigned deadline will not be eligible for full credit. Please communicate with me if you are having problems keeping to the schedule. It is better to talk through problems than ignore them.

ABSENCES:

The only absences that will be considered excused are for religious holidays or extenuating circumstances due to an emergency. By UT Austin policy, you must notify the instructor at least 14 days prior to any absence for observance of a religious holiday.

STUDENT RIGHTS & RESPONSABILITIES:

- You have a right to a learning environment that supports mental and physical wellness;
- You have a right to respect;
- You have a right to be assessed and graded fairly;
- You have a right to freedom of opinion and expression;
- You have a right to privacy and confidentiality;
- You have a right to meaningful and equal participation, to self-organize groups to improve your learning environment;
- You have a right to learn in an environment that is welcoming to all people. No student shall be isolated, excluded or diminished in any way.

With these rights come responsibilities:

- You are responsible for taking care of yourself, managing your time, and communicating with the teaching team and with others if things start to feel out of control or overwhelming;
- You are responsible for acting in a way that is worthy of respect and always respectful of others. Your experience with this course is directly related to the quality of the energy that you bring to it, and your energy shapes the quality of your peers' experiences;
- You are responsible for creating an inclusive environment and for speaking up when someone is excluded;
- You are responsible for holding yourself accountable to these standards, holding each other to these standards, and holding the teaching team accountable as well.

Personal pronoun preference:

Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with differences of race, culture, religion, politics, sexual orientation, gender, gender variance, and nationalities. Class rosters are provided to the instructor with the student's legal name. I will gladly honor your request to address you by an alternate name or gender pronoun. Please advise me of this preference early in the semester so that I may make appropriate changes to my records. Please forgive me if I make mistakes in addressing you.

ACADEMIC INTEGRITY:

Each student in the course is expected to abide by the University of Texas Honor Code: "As a student of The University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity." **Plagiarism is taken very seriously at UT.** Therefore, if you use words or ideas that are not your own (or that you have used in previous class), you must cite your sources. Otherwise you will be guilty of plagiarism and subject to academic disciplinary action, including failure of the course. You are responsible for understanding UT's Academic Honesty and the University Honor Code which can be found at the following web address:

<https://catalog.utexas.edu/law/academic-policies-and-procedures/honor-system/#universitycodeofconduct>

COURSE-INSTRUCTOR SURVEY:

The course and instructor will be evaluated using standard evaluation forms at the end of the course. Students are encouraged to provide continuous feedback to the instructor throughout the course. The most effective way to do this is by visiting instructor during office hours.

UNIVERSITY RESOURCES FOR STUDENTS:

Your success in this class is important. We will all need accommodations because we all learn differently. If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible. Together we'll develop strategies to meet both your needs and the requirements of the course. There are also a range of resources on campus:

Services for Students with Disabilities

The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Services for Students with Disabilities (SSD). Please refer to SSD's website for contact and more information: <http://diversity.utexas.edu/disability/>. If you are already registered with SSD, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.

Counseling and Mental Health Center

The Counseling and Mental Health Center (<https://cmhc.utexas.edu/>) serves UT's diverse campus community by providing high quality, innovative and culturally informed mental health programs and services that enhance and support students' wellbeing, academic and life goals. To learn more about your counseling and mental health options, call CMHC at 512-471-3515. If you are experiencing a mental health crisis, call the CMHC Crisis Line 24/7 at 512-471-2255.

Undergraduate Writing Center: <http://uwc.utexas.edu>

Libraries: <http://www.lib.utexas.edu>

ITS: <http://www.utexas.edu/its>

Student Emergency Services: <http://deanofstudents.utexas.edu/emergency>

BeVocal

BeVocal is a university-wide initiative to promote the idea that individual Longhorns have the power to prevent high-risk behavior and harm. At UT-Austin all Longhorns have the power to intervene and reduce harm. To learn more about BeVocal go to: <https://wellnessnetwork.utexas.edu/BeVocal>

IMPORTANT SAFETY INFORMATION:

If you have concerns about the safety or behavior of fellow students, TAs or Professors, call BCAL (the Behavior Concerns Advice Line): 512-232-5050. Your call can be anonymous. If something doesn't feel right – it probably isn't. Trust your instincts and share your concerns.

The following recommendations regarding emergency evacuation from the Office of Campus Safety and Security, 512-471-5767, <http://www.utexas.edu/safety>

Occupants of buildings on The University of Texas at Austin campus are required to evacuate buildings when a fire alarm is activated. Alarm activation or announcement requires exiting and assembling outside.

- Familiarize yourself with all exit doors of each classroom and building you may occupy. Remember that the nearest exit door may not be the one you used when entering the building;
- Students requiring assistance in evacuation shall inform their instructor in writing during the first week of class;
- In the event of an evacuation, follow the instruction of faculty or class instructors. Do not re-enter a building unless given instructions by the following: Austin Fire Department, The University of Texas at Austin Police Department, or Fire Prevention Services office.
- Link to information regarding emergency evacuation routes and emergency procedures can be found at: www.utexas.edu/emergency

TITLE IX REPORTING:

Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, unprofessional or inappropriate conduct of a sexual nature, dating/domestic violence and stalking at federally funded educational institutions. UT-Austin is committed to fostering a learning and working environment free from discrimination in all its forms. When unprofessional or inappropriate conduct of a sexual nature occurs in our community, the university can:

1. Intervene to prevent harmful behavior from continuing or escalating;
2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation;
3. Investigate and discipline violations of the university's [relevant policies](#).

Beginning January 1, 2020, Texas Senate Bill 212 requires all employees of Texas universities, including faculty, report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. I am a Responsible Employee and must report any Title IX related incidents that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a responsible employee. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, visit <http://www.titleix.utexas.edu>, contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.

Although graduate teaching and research assistants are not subject to Texas Senate Bill 212, they are still mandatory reporters under Federal Title IX laws and are required to report a wide range of behaviors we refer to as unprofessional or inappropriate conduct of a sexual nature, including the types of conduct covered under Texas Senate Bill 212. The Title IX office has developed supportive ways to respond to a survivor and compiled campus resources to support survivors.

OTHER ITEMS:

- Web-based, password-protected class sites are associated with all academic courses taught at the University. Students who do not want their names included in these electronic class rosters must restrict their directory information in the Office of the Registrar, Main Building, Room 1. For information on restricting directory information, see: <http://registrar.utexas.edu/students/records/restrictmyinfo>.
- Engineers are required to meet the highest levels of professional conduct and to responsibly handle complicated and often dangerous equipment and technical tasks. You will be held to those same expectations by your instructor during your studies and training. You are expected to obtain all pertinent safety training and to follow all safety procedures for equipment you handle and tasks you conduct. Failure to do so will result in a penalty on your course grade, a penalty completely at the discretion of the instructor. For example, explosive launching of any projectile in an inappropriate location or direction will result in a failing grade in this course. This is set as course policy so that it will be administered by the University for the instructor posthumously, if required.

Sharing of Course Materials is Prohibited:

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.

Class Recordings:

Any class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.